

K888 — the FK turn, taken to its verdict (Casey’s “please do”): FORM derived, VALUES structural. The definitive Gaunt/parity computation: a pure odd ($\ell=1$) condensate gives eigenvalues $\{0, +m, -m\} \rightarrow$ one massless + two degenerate (fails badly), so the condensate must carry many harmonics = a zonal profile with θ -width; the width is the one free/sourced parameter, and no natural profile (concentration, Szegő-kernel — K887) reproduces 1:207:3477, defeated by a model-independent obstruction (the log-gaps are unequal/decreasing, monotone profiles give increasing). A free profile over-fits three modes trivially (a fit, not a derivation). VERDICT: the lepton mass VALUES are IDENTIFIED, not derived — $(24/\pi^2)^6$ stays a T190 coincidence. What IS derived (banked): the FORM — one Toeplitz operator on D_{IV}^5 with a singular zonal condensate, exactly 3 modes (why-3), the hierarchy EXISTS (singular boundary measure), no-4th, the mixing pattern. NOT the K859 over-negative: the obstruction is understood, I tried the natural profiles, and the door stays open — a forced profile that fits would be a positive result to EXHIBIT (not fish for). Absent it, values structural, stated plainly.

Keeper | 2026-07-24 Fri | The FK turn, done to the end. Honest verdict, door left open. Plain.

The definitive computation (actual Gaunt/parity structure)

1. Pure odd ($\ell=1$) condensate — the minimal nearest-neighbor coupler: by parity (mode_0 even, mode_1 odd, mode_2 even; x odd), all diagonal entries vanish and $\langle 0|x|2\rangle=0$, so $M = \begin{bmatrix} 0 & a & 0 \\ a & 0 & b \\ 0 & b & 0 \end{bmatrix} \rightarrow$ eigenvalues $\{0, +\sqrt{(a^2+b^2)}, -\sqrt{(a^2+b^2)}\} \rightarrow$ one MASSLESS + two DEGENERATE. Nowhere near 1:207:3477. A pure odd condensate cannot work — you need EVEN harmonics ($\ell=0,2$) for the diagonal (tau mass).
2. So the condensate carries many harmonics = a zonal profile with θ -width. Rank floor (Lyra F686) precise: a single-latitude delta $\rightarrow M_{ij} = C_i(\theta)C_j(\theta) \cdot w \rightarrow$ rank-1 $\rightarrow$ one mass; a θ -WIDTH (spread over latitudes) $\rightarrow$ rank-3. The width is the one free/sourced parameter.
3. No natural width fits (K887): concentration $\kappa \approx 9.3 \rightarrow 1:86:5816$; Szegő-kernel $\rightarrow 1:121:4756$ — both miss. Obstruction (model-independent): the log-gaps are UNEQUAL/decreasing ($e \rightarrow \mu \sim 1.9 \times \mu \rightarrow \tau$); monotone peaked profiles give INCREASING gaps — the opposite.
4. A free profile over-fits three modes trivially $\rightarrow$ a fit, not a derivation.

★ The verdict

- DERIVED / banked (form + structure, value-independent): the mass sector = ONE Toeplitz operator on D_{IV}^5 with a singular zonal condensate; exactly 3 generations (why-3, T2525); the hierarchy EXISTS (singular boundary measure); no 4th; the mixing pattern (CKM small / PMNS large, F684). All sourced, all banked.
- STRUCTURAL / NOT derived (the values): 1:207:3477. No forced/natural condensate profile reproduces them; the unequal-gap shape defeats every simple profile; fitting requires a tuned width. $\frac{?}{?}$ the lepton mass VALUES are IDENTIFIED — $(24/\pi^2)^6 =$ T190 coincidence — not derived.

Why this is NOT the K859 over-negative (honest self-check)

- The FORM is fully derived and banked (not “dead” — a large, real result).

- The obstruction is understood (unequal gaps vs monotone profiles — a structural reason, not a failed search).
- I tried the natural profiles (concentration + Szegő-kernel), not one guess.
- The door stays open: a *forced* profile that reproduces the unequal gaps would flip the verdict to derived — but it must be EXHIBITED (a positive result), not fished for by scanning profiles until one fits. Absent such a forced profile, the values are structural.
- So: Casey, if you have a forced-profile reading (as you did with the climb on K859), it's testable — exhibit the profile, and the diagonalization decides. Absent it, structural.

The honest close of the two-day arc

- The lepton mass VALUES: structural (identified). The FK turn, done to the end, doesn't derive them — no natural forced profile, understood obstruction.
- Everything structural is banked: why-3 (T2525), hierarchy-exists (singular measure), the one-Toeplitz framework, the mixing pattern, the EW sector. These are the durable, real results.
- The forward work (unblocked): the quark structural lane + consolidating the durable wins into papers. The lepton VALUES rest at identified/structural, honestly.
- Nothing false banked across the whole arc; the discipline held in all directions including on the auditor (and this verdict itself keeps the door open rather than over-claiming a negative).

Directions

- ★ TEAM: the lepton mass VALUES are structural (K888) — bank the FORM (T2525, framework, hierarchy, mixing pattern), tier the VALUES as identified (T190). Forward: the quark structural lane + the durable papers.
- ★ IF a forced condensate profile is proposed (Casey or Grace) that reproduces the unequal-gap shape from the geometry (no tuning) → test it (the diagonalization decides). That's the only path from structural → derived, and it must be exhibited.
- KEEPER: FK turn verdict (K888) — form derived, values structural; obstruction understood; door open for a forced profile; not the K859 over-negative.

— Keeper K888, 2026-07-24. FK turn VERDICT: FORM derived (one Toeplitz on D_{IV}^5 , singular zonal condensate, 3 modes/why-3, hierarchy-exists, no-4th, mixing pattern — banked), VALUES structural. Pure odd $\ell=1 \rightarrow \{0, m, m\}$ (fails); need θ -width profile; no natural width fits (K887), obstruction = unequal/decreasing log-gaps (monotone profiles give opposite); free profile over-fits (fit). Lepton VALUES = IDENTIFIED (T190 $(24/\pi^2)^6$), not derived. NOT the K859 over-negative: form banked, obstruction understood, natural profiles tried, door OPEN — a forced profile reproducing the unequal gaps would flip to derived but must be EXHIBITED not fished. See [[Keeper_K887_ranging_shot_no_simple_condensate_value_reproduces_the_masses_the_obstruction_is_UNEQUAL_log_gaps_07-24]], [[Keeper_K886_where_we_go_the_lepton_value_reduces_to_ONE_number_the_condensate_latitude_theta_rank_floor_07-24]], Lyra F686 (rank floor), T190 $((24/\pi^2)^6$ identified), T2525 (why-3 banked).